Hydraulic Pressure Testing Machine

Category: **Plumbing & Pipe Fitting** Type: **Full SOP** Priority: **High** 4 labs

Covers these lab items: Hydraulic Testing Machine; Hydraulic water pressure Machine

What it is used for

A hydraulic pressure testing machine applies controlled water pressure to pipes, fittings, or other approved components to check for leakage and pressure integrity. Correct setup, gradual pressurisation, and controlled depressurisation help produce reliable results and reduce the risk of sudden failure.

Main parts

- Water reservoir or supply connection provides the test liquid.
- Test pump raises the water pressure.
- Pressure gauge or digital sensor displays the applied pressure.
- Pressure regulator or control valve adjusts the pressure.
- Test hoses and fittings connect the pump to the test item.
- Isolation valves control water flow through the test setup.
- Relief valve or protective device limits excessive pressure where fitted.
- Frame and mounting fixtures support the test equipment and component.

Before you start

1. Only trained personnel should conduct tests using the approved procedure and rated equipment.
2. Read the machine manual and identify the test item's specified pressure and hold time.
3. Inspect hoses, couplings, gauges, valves, seals, and the test item for damage.
4. Confirm the machine, hoses, fittings, and test item are rated for the planned pressure.
5. Secure the test item and verify all connections are correctly fitted.
6. Use water or the specified compatible test liquid and remove trapped air where the procedure requires it.

How to use

1. Switch off the pump and release any residual pressure before connecting the test item.
2. Connect the test item securely using the correct fittings and approved test arrangement.
3. Fill the test circuit with the specified liquid and vent trapped air according to the procedure.
4. Close the required valves and check that all connections are secure.
5. Start the pump and raise pressure gradually to the specified test value.
6. Observe the gauge and inspect the test setup from a safe position for leakage or abnormal movement.
7. Maintain the specified pressure for the approved hold time and record readings.

8. If pressure drops or leakage appears, stop the test and follow the approved depressurisation procedure.
9. Release pressure slowly through the designated valve and confirm the gauge reads zero before disconnecting.
10. Drain the test item as required, inspect it, and record the result and any faults.

⚠ Safety

- Never exceed the rated pressure of the machine, hoses, fittings, or test item.
- Never loosen a connection while the system is pressurised.
- Keep people clear of hose ends, fittings, and the line of possible component failure.
- Do not use damaged hoses, gauges, valves, or test fixtures.
- Use water for hydrostatic testing unless the approved procedure specifies another compatible liquid; never substitute compressed gas without a specifically engineered procedure.
- If a leak, bulging hose, unusual noise, or sudden pressure change occurs, stop the pump and depressurise from a safe position.

Care & storage

- Drain and dry the equipment as recommended to reduce corrosion and contamination.
- Inspect hoses, seals, couplings, and valves after each test.
- Keep pressure gauges and sensors within their calibration or verification schedule.
- Store the machine and accessories in a clean, dry location and record maintenance.

Common mistakes to avoid

- Increasing pressure too quickly or exceeding the specified test limit.
- Leaving trapped air in the circuit when the procedure requires venting.
- Attempting to tighten or disconnect fittings while pressure remains.
- Failing to record the test pressure, hold time, and observed result.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.